

Category-Aware Explainable Machine Learning for Urban Service Demand Forecasting in Smart Cities

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Abstract

Urban 311 systems record large-scale traces of reported municipal demand, but forecasting operationally useful surges remains difficult because reports are spatially uneven, service categories differ, and reporting behavior shapes observed counts. We formulate next-week abnormal reported-demand forecasting as a neighborhood-week-service-category early-warning task. Using New York City, the analysis integrates 30.9 million NTA-matched 311 requests with shifted historical demand features, calendar timing, NOAA weather exposure, OpenStreetMap point-of-interest context, and PLUTO built-environment descriptors. A target-week chronological protocol trains on 2015-2022 target weeks, validates on 2023, and tests on 2024-2025. The validation-selected decision strategy averages LightGBM and XGBoost scores and applies complaint-category-specific thresholds selected only on validation. On the held-out future test period, it obtains $F1 = 0.3839$, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, and ROC-AUC = 0.7720. SHAP analysis shows that historical temporal demand features dominate, while service category, built environment, POI context, weather, and calendar features add interpretable signal. The system is positioned as decision support for prioritizing elevated reported-demand risk, not as a causal model of all urban conditions.

Keywords: urban computing, smart cities, 311 service requests, service-demand forecasting, explainable machine learning, SHAP, spatiotemporal modeling

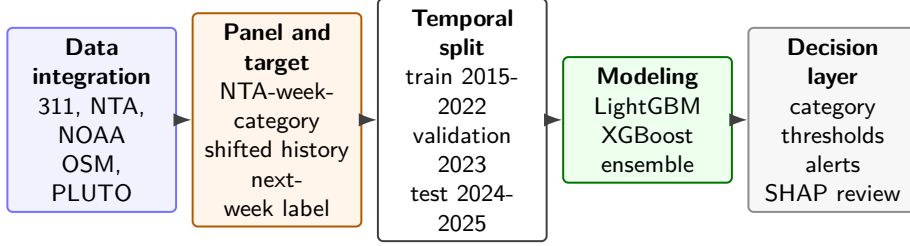


Fig. 1 Overview of the leakage-controlled temporal pipeline. The modeling target is next-week abnormal reported demand for each NTA, week, and service category; OSM and PLUTO are used as static context snapshots.

1 Introduction

Urban service systems increasingly rely on administrative and citizen-generated data to monitor conditions, allocate attention, and anticipate operational pressure. NYC 311 is especially valuable because it records requests across infrastructure, housing, sanitation, public safety, environmental, and quality-of-life domains. Yet 311 counts are not direct measurements of all underlying urban problems. They reflect both conditions and the propensity to report them, which varies with access, awareness, trust, service expectations, and governance processes [1–4]. Forecasts built from these data therefore need careful wording: the target is abnormal reported service demand, not the true incidence of every urban issue.

We study a prospective early-warning task at the neighborhood tabulation area (NTA), week, and complaint-category level. For each feature week, the model estimates whether the following week will show abnormal reported demand relative to recent local history. The design combines historical 311 demand, calendar timing, weekly weather exposure, OpenStreetMap (OSM) point-of-interest context, and PLUTO built-environment descriptors. The target and temporal features are leakage-controlled: lags and rolling statistics are shifted so only information available before the prediction week is used. Static OSM and PLUTO snapshots are treated as contextual descriptors and are discussed as a snapshot/look-ahead limitation rather than as dynamic covariates.

Three questions guide the analysis. First, can a multi-source NTA-week-category panel forecast next-week abnormal reported demand under a future chronological test? Second, how does a complaint-category-specific decision layer change precision, recall, and alert-volume trade-offs across service domains? Third, which temporal and urban-context feature groups drive model predictions? The contributions are a category-aware early-warning formulation for urban service demand, a large NYC panel built from 30,940,129 NTA-matched 311 requests and public contextual data, a target-week chronological evaluation with validation-only threshold selection, and SHAP-based interpretation of model behavior. Figure 1 summarizes the workflow.

2 Related Work

Urban computing links administrative, geographic, and sensed data to support city analysis and decision making [5]. Recent spatiotemporal learning surveys emphasize

the value of temporal structure, spatial context, and prospective validation in urban prediction problems [6]. Within 311 research, prior work shows that complaint structure can distinguish urban locations [1], while reporting bias can distort data-driven governance decisions [2, 3]. Related domain studies connect 311 requests to flooding [7], extreme heat [8], foodborne illness signals [9], open-data curation [4], and municipal service-level design [10]. Our task differs by forecasting future abnormal reported demand across multiple service categories in a single citywide panel.

Contextual urban features can improve service-demand models when interpreted carefully. OSM provides volunteered geographic information and POI records useful for describing neighborhood function, but coverage and tagging are uneven [11, 12]. POI and land-use features have been used for urban change, functional-zone, land-use, and traffic-flow modeling [13–16]. In this work, OSM POIs and PLUTO built-environment summaries support predictive context; they are not interpreted as contemporaneous causal drivers.

The forecasting target is an imbalanced binary event, making ordinary accuracy uninformative. Imbalance-learning surveys and evaluation work motivate precision, recall, F1, PR-AUC, ROC-AUC, and balanced accuracy as complementary views [17–21]. Gradient-boosted tree methods such as XGBoost and LightGBM are strong tabular baselines for nonlinear interactions [22, 23]. Because operational use requires transparency, we use SHAP and TreeSHAP to explain a tuned LightGBM component while following cautions that explanations of black-box models should not be mistaken for causal guarantees [24–26].

3 Methodology

3.1 Data and Task Formulation

The project integrates five public data sources: NYC Open Data 311 Service Requests [27], 2020 NTA boundaries [28], NOAA GHCN-Daily weather for Central Park station GHCND:USW00094728 [29], OSM POIs queried through Overpass [30], and NYC PLUTO/MapPLUTO land-use records [31]. Weekly 311 counts use Monday week starts and are spatially joined to 2020 NTAs. Weather covariates summarize the feature week t , not the target week $t + 1$. OSM and PLUTO variables are fixed extraction snapshots from 2026-06-25 and are used as static context descriptors.

The analytical unit is (n, t, c) , where n is an NTA, t is a feature-week start date, and c is one of nine stable complaint categories: noise, housing, water/sewer, public safety, parking/traffic, environment, infrastructure, sanitation, and other. These categories reduce sparsity while preserving service meaning. The dense panel contains 262 NTAs, 575 weekly periods, and nine categories, yielding 1,355,850 rows. Relative to the full dense panel, 30,654 rows are excluded due to warm-up/history requirements and unavailable next-week targets, leaving 1,325,196 model-ready rows. Table 1 summarizes the construction.

For each feature week t , all predictors are available after that week closes. The target week is $t + 1$, seven days after the feature-week start date:

$$y_{n,t,c} = \mathbf{1}\{\text{count}_{n,t+1,c} > \mu_{n,t,c}^{sw} + 1.5\sigma_{n,t,c}^{sw}\}.$$

Table 1 Dataset construction summary.

Item	Value	Item	Value
NTA-matched 311 Unit	30,940,129	Match rate	99.9777%
Valid-label rows	NTA-week-category 1,334,628	Panel rows	1,355,850
Excluded rows	30,654	Model-ready rows	1,325,196
Weeks	575	NTAs	262
Coverage	2014-12-29-2025-12-29	Categories	9
OSM POIs	77,083	NOAA days	4,018
Predefined schema	215 inputs	Output columns	252
		Final feature subset	213 inputs

Table 2 Compact tuning and final-score protocol for gradient-boosted models.

Item	Setting
Feature set	213-input no-COVID-period subset; one-hot expansion to 229 model columns
Tuning sample/seed	Stratified train sample, <code>max_train_rows</code> = 300,000, seed 42; selected candidates rebuilt on all 954,990 training rows
LightGBM grid	Depth/leaf/lr: (8,63,0.05), (6,31,0.03), (10,127,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
XGBoost grid	Depth/lr: (8,0.05), (6,0.03), (5,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
Selection	Validation F1; selected LightGBM depth 6, 31 leaves, lr 0.03, sqrt weight; selected XGBoost depth 6, lr 0.03, sqrt weight
Thresholds	Validation grid 0.01–0.99 step 0.005 plus score quantiles; category fallback to global threshold if validation positives < 30

The rolling mean $\mu_{n,t,c}^{8w}$ and standard deviation $\sigma_{n,t,c}^{8w}$ are computed from a shifted historical eight-week window. The label therefore identifies next-week abnormal reported demand relative to each local recent baseline.

3.2 Predictive Models and Category-Aware Decision Layer

The model-ready configuration starts from a predefined 215-input schema specified before model fitting. The final tuned comparison uses a 213-input no-COVID-period subset, excluding two period indicators so performance does not rely on COVID-era labels; one-hot encoding expands this to 229 model columns. Historical features include lag counts, rolling statistics, ratios to historical baselines, current demand, and history-availability flags. Calendar, weather, OSM, and PLUTO features add temporal and neighborhood context.

We compare rule baselines, logistic regression, LightGBM, XGBoost, and an equal-weight LightGBM/XGBoost ensemble. Numeric medians and categorical levels are fitted on the training period only. Hyperparameters are selected from stratified 300,000-row tuning runs for reproducibility, then final scores are rebuilt on all 954,990 training rows. Table 2 records the compact protocol.

Let $s_{n,t,c}^L$ be the tuned LightGBM score and $s_{n,t,c}^X$ the tuned XGBoost score. The ensemble score is $s_{n,t,c}^E = 0.5s_{n,t,c}^L + 0.5s_{n,t,c}^X$. The category-aware decision layer assigns each category c a validation-selected threshold τ_c , producing $\hat{y}_{n,t,c} = \mathbf{1}\{s_{n,t,c}^E \geq \tau_c\}$.

This is category-aware at the decision stage: the same probability score can map to different alert decisions because service domains differ in score distribution, prevalence, and operating cost.

3.3 Evaluation and Explainability Protocol

Evaluation uses a chronological split by target week: 2015-2022 for training, 2023 for validation, and 2024-2025 for testing. The model-ready split contains 954,990 train rows with 121,306 positives, 122,616 validation rows with 15,574 positives, and 247,590 test rows with 31,570 positives. Validation and test positive shares are approximately 12.7%. All thresholds are selected on validation only and evaluated unchanged on the held-out test period.

F1 is the primary validation-ranking metric because the alerting task requires balancing event detection and alert conversion. We also report precision, recall, PR-AUC, ROC-AUC, balanced accuracy, alert rate, and confusion matrices. Threshold tuning is treated as a method component because it changes hard-alert metrics and alert volume; it does not change PR-AUC or ROC-AUC for a fixed score vector.

For explainability, SHAP is computed for the tuned LightGBM component. Since the final deployed score is an ensemble with category-specific thresholds, SHAP values are interpreted as evidence about a representative tree component under the same feature design and chronological protocol, not as a direct explanation of every final alert or as causal attribution.

4 Results

4.1 Overall Performance and Ablation Results

Rule and logistic baselines confirm that simple history alone is not enough for the final alerting task. On the 2024-2025 target-week test period, current-week persistence obtains $F1 = 0.3111$ and $PR-AUC = 0.2359$, a seasonal lag-52 margin obtains $F1 = 0.2423$ and $PR-AUC = 0.1773$, and the strongest rule baseline, a rolling z-score rule, obtains $F1 = 0.3332$ and $PR-AUC = 0.2658$. Logistic regression with historical features obtains $F1 = 0.3256$ and $PR-AUC = 0.2598$. LightGBM ablations show ranking quality rising from $PR-AUC = 0.2770$ with historical features alone to 0.3158 with full context.

Table 3 compares final decision strategies. Following the validation-selection rule, the operationally preferred category-aware strategy is the LightGBM/XGBoost ensemble with complaint-category-specific thresholds. On the held-out future test period, it achieves $F1 = 0.3839$, precision = 0.3071, recall = 0.5117, $PR-AUC = 0.3365$, $ROC-AUC = 0.7720$, and balanced accuracy = 0.6715. The global-threshold ensemble has slightly higher test F1, but it applies one alert rule across service domains; the category-specific strategy is retained as the validation-selected and operationally interpretable decision layer.

The final test confusion matrix contains 16,154 true positives, 36,440 false positives, 15,416 false negatives, and 179,580 true negatives. The model captures about half of true next-week abnormal reported-demand events while roughly three out of

Table 3 Final decision-layer comparison. Thresholds are selected on validation and evaluated unchanged on the future test period.

Model	Threshold	Val F1	Test F1	Prec.	Rec.	PR-AUC	ROC-AUC	Alert rate
Ensemble LGBM(0.50)+XGB(0.50)	Per category	0.3809	0.3839	0.3071	0.5117	0.3365	0.7720	0.2124
Tuned XGBoost	Per category	0.3806	0.3845	0.3058	0.5175	0.3367	0.7722	0.2158
Tuned LightGBM	Per category	0.3802	0.3836	0.3022	0.5249	0.3346	0.7709	0.2214
Ensemble LGBM(0.50)+XGB(0.50)	Global	0.3777	0.3865	0.3001	0.5426	0.3365	0.7720	0.2305

Table 4 Final model test performance by complaint category on the 2024-2025 target-week period.

Category	Positives	Alerts	PR-AUC	F1	Precision	Recall
Noise	3,982	5,729	0.4229	0.4578	0.3880	0.5583
Public safety	3,536	6,482	0.3312	0.4097	0.3166	0.5803
Housing	3,720	5,412	0.3860	0.4025	0.3396	0.4941
Water/sewer	3,462	6,090	0.3642	0.3938	0.3089	0.5433
Other	3,620	6,695	0.3237	0.3762	0.2898	0.5359
Parking/traffic	3,586	5,811	0.3255	0.3708	0.2998	0.4858
Environment	2,896	5,924	0.3105	0.3633	0.2704	0.5532
Infrastructure	3,596	5,900	0.2764	0.3484	0.2803	0.4600
Sanitation	3,172	4,551	0.2570	0.3165	0.2685	0.3852

ten alerts correspond to true events. The top 1% of scored test rows has precision 0.5965 and 4.68-times lift over the test base rate; the top 5% has precision 0.4548; and the top 10% has precision 0.3866. These concentration results position the model as prioritization support rather than an automatic allocation rule.

4.2 Category-Level Alert Analysis

Performance varies across municipal service domains (Table 4). Noise has the strongest F1, followed by public safety, housing, and water/sewer. Sanitation, infrastructure, and environment are weaker by F1, although several retain moderate recall. Positives in Table 4 are observed abnormal next-week rows within the held-out test set; alerts are rows whose ensemble score exceeds the validation-selected threshold for that complaint category. Alert counts therefore need not be proportional to positives. They depend on the category threshold, the category score distribution, separation between positive and negative rows, and the validation operating point. The counts are consistent with the test confusion matrix: summing the category positives gives 31,570 and summing the category alerts gives 52,594.

Figure 2 makes the operating-point trade-off visible. Noise has more observed positives than public safety but fewer alerts; its selected operating point yields higher precision and slightly lower recall, while public safety issues more alerts to capture higher recall at lower precision. This is not a data contradiction, but a consequence of category thresholds and score distributions. Relative to a global ensemble threshold, the category-specific ensemble has macro-F1 0.3821 versus 0.3838, macro precision 0.3069 versus 0.2989, macro recall 0.5107 versus 0.5394, and mean alert rate 0.2124 versus 0.2305. The result should be read as a validation-selected, operationally preferred category-aware strategy, not as the best aggregate-F1 strategy.

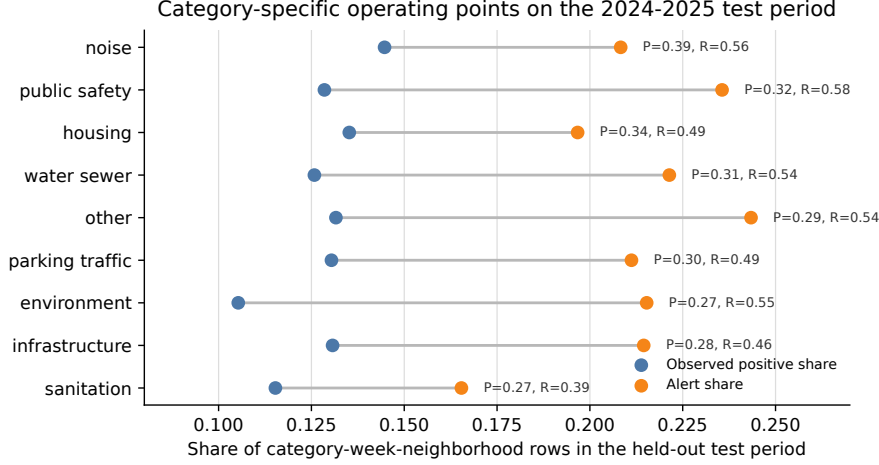


Fig. 2 Observed positive share and validation-threshold alert share by complaint category in the held-out test period. Labels show category precision and recall at the selected operating point.

4.3 Explanation Results

Figure 3 summarizes SHAP importance by feature group for the tuned LightGBM component. Historical temporal features dominate, accounting for 58.96% of summed mean absolute SHAP importance across 21 features. Service category accounts for 10.73% across nine indicators, PLUTO built environment for 10.18% across 53 features, OSM POI context for 8.13% across 103 features, weather for 4.88% across 22 features, calendar features for 4.77% across four features, and current demand for 1.61% for one feature. Because these are summed group importances, groups with more features can accumulate larger total contribution. The leading individual features include `rolling_8w_std`, `ratio_to_8w_mean`, `week_of_year`, `lag_52w_count`, `rolling_4w_mean`, `lag_1w_count`, service-category indicators, and `current_complaint_count`.

Two local examples illustrate how explanations should be used. A true-positive housing alert in Inwood for feature week 2025-10-06 has a LightGBM score of 0.954 and is associated with eight-week volatility, seasonal timing, and residential built-environment features. A false-positive public-safety alert in the Lower East Side for feature week 2025-06-30 has a LightGBM score of 0.940; annual and recent lag counts and current reported demand push the score upward, while eight-week volatility and zero weekly precipitation push it downward. These cases support analyst review but do not establish causal effects.

5 Discussion

5.1 Operational and Methodological Implications

The model is most useful as early-warning decision support. Recall of 0.5117 means it captures about half of future abnormal reported-demand events, while precision

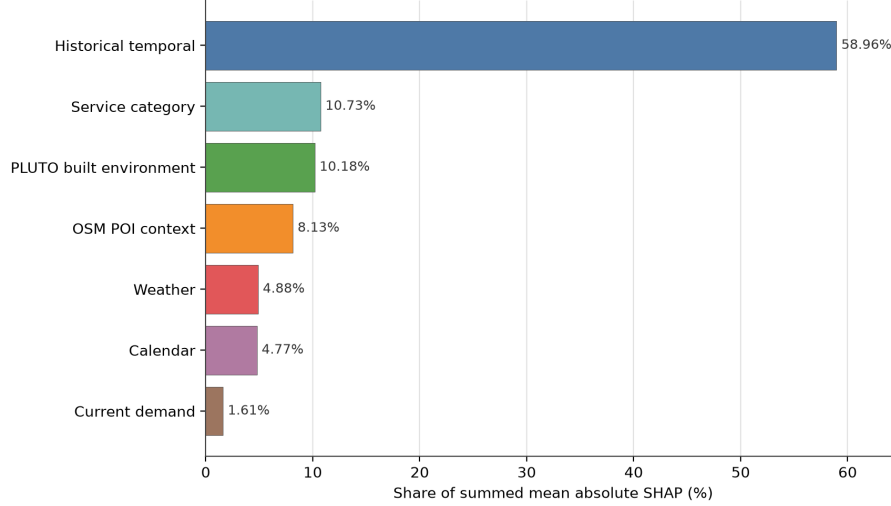


Fig. 3 Normalized summed mean absolute SHAP importance for the tuned LightGBM explanation backbone. Historical temporal demand features dominate, while service-category, PLUTO, OSM POI, weather, and calendar features add smaller but interpretable contextual contributions.

of 0.3071 means many alerts are false positives. Agencies should therefore use alerts to prioritize review, pre-position attention, or monitor risk queues, with final actions incorporating severity, staffing, and local knowledge.

Category-specific thresholds make the alert layer more realistic because service domains do not share the same base rates, volatility, response costs, or tolerance for missed events. A sanitation alert and a public-safety alert can have different operational consequences even when their model scores are similar. The small aggregate-F1 difference between global and category-specific thresholds reinforces the main point: the benefit is explicit service-domain trade-off control, not a claim of universal metric dominance.

5.2 Limitations and Future Work

Several limits should guide interpretation. NYC 311 data measure reported service demand, not all underlying urban problems, and reporting propensity can reproduce socio-spatial inequities. OSM POI and PLUTO are static 2026 snapshots used across a 2015-2025 panel, so they carry snapshot/look-ahead risk and should be interpreted only as broad context. NOAA weather is weekly city-level exposure rather than NTA microclimate. Hyperparameters are selected from sampled tuning runs, although final reported scores are rebuilt on all training rows; three sampled seeds produced test F1 0.3817 ± 0.0021 , PR-AUC 0.3313 ± 0.0025 , and ROC-AUC 0.7678 ± 0.0007 . Future work should test transferability to other cities, alternative abnormal-demand definitions, fairness-aware reporting adjustments, repeated full-training runs, and links between alerts and operational response outcomes.

6 Conclusion

We present an explainable machine-learning pipeline for forecasting next-week abnormal reported demand in NYC 311 data at the neighborhood, week, and service-category level. The NTA-week-category formulation, multi-source context, and target-week chronological split frame 311 forecasting as an early-warning task rather than generic count prediction. The validation-selected LightGBM/XGBoost ensemble with category-specific thresholds achieves future-test $F1 = 0.3839$, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, and ROC-AUC = 0.7720. SHAP shows that historical temporal dynamics dominate prediction, while built environment, POI context, weather, service category, and calendar features add smaller but interpretable contextual contributions. The findings support category-aware decision layers as a practical way to make service-domain alert trade-offs explicit while keeping the central limitation clear: 311 data forecast reported demand, not the complete distribution of urban need.

Declarations

Data and code availability. Source data are NYC Open Data 311 ([erm2-nwe9](#)), NOAA GHCN-Daily station [GHCND:USW00094728](#), OpenStreetMap Overpass, NYC PLUTO/MapPLUTO ([64uk-42ks](#)), and 2020 NTA boundaries ([9nt8-h7nd](#)); the local snapshot was extracted on 2026-06-25. Scripts, mappings, configurations, and evaluation code are available at https://github.com/PhongLam26/SIMC_NYC; processed data release is subject to licensing, redistribution, and storage constraints.

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Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le served as academic advisor and supervised the study design, methodology, and manuscript editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed validation and review.

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